

# ALK fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes ALK gene fusions from the msk\_impact\_50k\_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ( $p=0.00191661$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for ALK.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

### Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain,  $p=0.00191661$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

### Domain disruption

Domain-disruption analysis was skipped: ALK has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected  $p=0.000999001$ , statistically significant at  $\alpha=0.05$ ). This is 12 aa from the nearest configured domain boundary at 960 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared `Domain_retention_flags` groups "retained" ( $n=261$ ) and "not\_retained" ( $n=10$ ). For `Frame_status`, retained: 222/227 (97.8%, 95% CI 94.9%-99.1%); not\_retained: 4/5 (80.0%, 95% CI 37.6%-96.4%). Welch's t-test comparing `Tumor_variant_count` between groups gave means of 65.6139 vs 31.8,  $p=0.00432792$  (statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

### Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

## **Joint partner**

Joint partner reported: ALK has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| EML4         | 221         | 0.8125               | 0.30004340774793187    | None                    | 0.8865878486549185       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.6503541781991878  | 1    |
| LRP1B        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 1.0                      | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3683101330469789  | 2    |
| EMILIN1      | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.9559412550066756       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.36004911836073056 | 3    |
| CLTC         | 2           | 0.007352941176470588 | 0.30004340774793187    | None                    | 0.931909212283044        | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3569217868206379  | 4    |
| FN1          | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.931909212283044        | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.35554311035004965 | 5    |
| NOTCH1       | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.931909212283044        | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.35554311035004965 | 6    |
| NRP2         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.931909212283044        | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.35554311035004965 | 7    |
| STRN         | 7           | 0.025735294117647058 | 0.30004340774793187    | None                    | 0.8865153538050734       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.35530382070895955 | 8    |
| KIF5B        | 5           | 0.01838235294117647  | 0.30004340774793187    | None                    | 0.8945260347129506       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3540484704380101  | 9    |
| PLEKHA7      | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.9158878504672897       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3525391050095957  | 10   |
| SPTBN1       | 4           | 0.014705882352941176 | 0.30004340774793187    | None                    | 0.8895193591455274       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.35173104229853    | 11   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| RANBP2       | 3           | 0.011029411764705883 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.34953878104823544 | 12   |
| PICALM       | 3           | 0.011029411764705883 | 0.30004340774793187    | None                    | 0.8811748998664887       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.348787779713122   | 13   |
| SQSTM1       | 2           | 0.007352941176470588 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.34816010457764723 | 14   |
| TPM1         | 2           | 0.007352941176470588 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.34816010457764723 | 15   |
| CTNNA1       | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 16   |
| DCTN1        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 17   |
| DIAPH2       | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 18   |
| DYSF         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 19   |
| GCC2         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 20   |
| HMBOX1       | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 21   |
| PLEKHH2      | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 22   |
| PPP1CB       | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 23   |
| TFG          | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059   | 24   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score    | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|--------------------|------|
| TRAF3        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059  | 25   |
| TTC27        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059  | 26   |
| VCL          | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059  | 27   |
| ZFPM2        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.8851802403204272       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.346781428107059  | 28   |
| H6PD         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.7503337783711616       | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3214977164915717 | 29   |
| CTSE         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.39118825100133514      | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2541579301097292 | 30   |
| SOS1         | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | None                     | 0.6853431332412958         | ['confidence_certainty', 'domain_retention', 'recurrence']                       | 0.2225355483655125 | 31   |
| ASAP2        | 1           | 0.003676470588235294 | 0.30004340774793187    | None                    | 0.0                      | 0.6853431332412958         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1808101330469789 | 32   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio           | p_value                | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|----------------------|------------------------|--------------------|
| 223      | 0                      | 261              | 1                      | 9                | 0.0                  | 0.03690036900369004    | 1.4329692908744058 |
| 516      | 0                      | 261              | 2                      | 8                | 0.0                  | 0.001230012300123001   | 2.9100905455940684 |
| 785      | 0                      | 261              | 3                      | 7                | 0.0                  | 3.658029145347215e-05  | 4.436752838604533  |
| 939      | 2                      | 259              | 3                      | 7                | 0.018018018018018018 | 0.0003515999093716071  | 3.4539512455419974 |
| 972      | 2                      | 259              | 4                      | 6                | 0.011583011583011582 | 1.3820564716971664e-05 | 4.8594742110534295 |
| 1023     | 10                     | 251              | 4                      | 6                | 0.05976095617529881  | 0.0007961806348620647  | 3.098988389769786  |
| 1028     | 11                     | 250              | 4                      | 6                | 0.066                | 0.0010657190672512412  | 2.9723572639982643 |
| 1032     | 12                     | 249              | 4                      | 6                | 0.07228915662650602  | 0.0013947454739763106  | 2.8555050392241372 |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value               | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|---------------------|-----------------------|--------------------|
| 1035     | 13                     | 248              | 4                      | 6                | 0.07862903225806452 | 0.0017901726849544938 | 2.747105073765603  |
| 1040     | 14                     | 247              | 4                      | 6                | 0.08502024291497975 | 0.002258991897970247  | 2.646085326703862  |
| 1042     | 16                     | 245              | 4                      | 6                | 0.09795918367346938 | 0.003444927806265043  | 2.4628198749665864 |
| 1045     | 18                     | 243              | 4                      | 6                | 0.11111111111111111 | 0.005008607520231548  | 2.300282998750876  |
| 1047     | 19                     | 242              | 4                      | 6                | 0.11776859504132231 | 0.005949364209574388  | 2.225529443519691  |
| 1051     | 20                     | 241              | 4                      | 6                | 0.12448132780082988 | 0.007005060123461933  | 2.154588132872754  |
| 1052     | 22                     | 239              | 4                      | 6                | 0.13807531380753138 | 0.00948736346530418   | 2.0228544610771384 |
| 1058     | 254                    | 7                | 10                     | 0                | 0.0                 | 1.0                   | -0.0               |
| 1059     | 255                    | 6                | 10                     | 0                | 0.0                 | 1.0                   | -0.0               |
| 1061     | 258                    | 3                | 10                     | 0                | 0.0                 | 1.0                   | -0.0               |
| 1064     | 259                    | 2                | 10                     | 0                | 0.0                 | 1.0                   | -0.0               |
| 1070     | 260                    | 1                | 10                     | 0                | 0.0                 | 1.0                   | -0.0               |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 222 | 39 |
| 4   | 6  |

*domain\_retention\_descriptives*

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 271                     | 261                  | 0               | 10               | 0.0                                             |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| ASAP2        | 1           |
| CLTC         | 2           |
| CTNNA1       | 1           |
| CTSE         | 1           |
| DCTN1        | 1           |
| DIAPH2       | 1           |
| DYSF         | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| EMILIN1      | 1           |
| EML4         | 221         |
| FN1          | 1           |
| GCC2         | 1           |
| H6PD         | 1           |
| HMBOX1       | 1           |
| KIF5B        | 5           |
| LRP1B        | 1           |
| NOTCH1       | 1           |
| NRP2         | 1           |
| PICALM       | 3           |
| PLEKHA7      | 1           |
| PLEKHH2      | 1           |
| PPP1CB       | 1           |
| RANBP2       | 3           |
| SOS1         | 1           |
| SPTBN1       | 4           |
| SQSTM1       | 2           |
| STRN         | 7           |
| TFG          | 1           |
| TPM1         | 2           |
| TRAF3        | 1           |
| TTC27        | 1           |
| VCL          | 1           |
| ZFPM2        | 1           |

## Per-event results (results.tsv)

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name      | part_ner_gene | fra_me_status   | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains              | lost_do_mains                             | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|-------------------|---------------|-----------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|-------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0024<br>679-T01-IM<br>6-39 | P-0024<br>679-T0<br>1-IM6 |              | ALK-<br>CTS<br>E  | CTS<br>E      | unk<br>now<br>n | five<br>_pr<br>ime   | 295435<br>94          | 7                | 516                          | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                     | ALK (NM_004304) -<br>CTSE(NM_001910)<br>Rearrangement: t(1;2)(q32.<br>1;p23.3)(chr1:g.206318648<br>; chr2:g.29543594)  <br>Antisense Fusion                                           | NA                            |
| EVT-P-0020<br>387-T01-IM<br>6-40 | P-0020<br>387-T0<br>1-IM6 |              | ALK-<br>DCT<br>N1 | DCT<br>N1     | in-fr<br>ame    | thre<br>e _pr<br>ime | 294468<br>96          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | DCTN1 (NM_004082) -<br>ALK (NM_004304)<br>rearrangement: c.3196+23<br>8:DCTN1_c.3173-502:ALK<br>del   Protein Fusion: in<br>frame {DCTN1:ALK}                                         | NA                            |
| EVT-P-0006<br>776-T01-IM<br>5-44 | P-0006<br>776-T0<br>1-IM5 |              | ALK-<br>DYS<br>F  | DYS<br>F      | unk<br>now<br>n | five<br>_pr<br>ime   | 294474<br>18          | 19               | 1058                         | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                     | ALK (NM_004304) - DYSF<br>(NM_001130987)<br>rearrangement: c.3172+90<br>9:ALK_c.3402+7348:DYSF<br>dup   Antisense fusion                                                              | NA                            |
| EVT-P-0031<br>827-T02-IM<br>6-45 | P-0031<br>827-T0<br>2-IM6 |              | ALK-<br>EML<br>4  | EML<br>4      | in-fr<br>ame    | thre<br>e _pr<br>ime | 294466<br>46          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-13 fused to<br>ALK exons 20-29): c.1490-<br>639:EML4_c.3173-252:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                            |
| EVT-P-0028<br>287-T01-IM<br>6-46 | P-0028<br>287-T0<br>1-IM6 |              | ALK-<br>EML<br>4  | EML<br>4      | in-fr<br>ame    | thre<br>e _pr<br>ime | 294466<br>61          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-5 fused<br>with ALK exons 20-29): c.4<br>94-2623:EML4_c.3173-267<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                            |
| EVT-P-0009<br>301-T01-IM<br>5-47 | P-0009<br>301-T0<br>1-IM5 |              | ALK-<br>EML<br>4  | EML<br>4      | in-fr<br>ame    | thre<br>e _pr<br>ime | 294466<br>51          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 with ALK<br>exons 20-29) : c.668-6693:<br>EML4_c.3173-257:ALKinv  <br>Protein fusion: in frame<br>{EML4-ALK}          | NA                            |
| EVT-P-0009<br>301-T02-IM<br>6-48 | P-0009<br>301-T0<br>2-IM6 |              | ALK-<br>EML<br>4  | EML<br>4      | in-fr<br>ame    | thre<br>e _pr<br>ime | 294466<br>51          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 fused to<br>ALK exons 20-29): c.668-6<br>693:EML4_c.3173-257:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0009<br>339-T01-IM<br>5-49 | P-0009<br>339-T0<br>1-IM5 |              | ALK-<br>EML<br>4  | EML<br>4      | in-fr<br>ame    | thre<br>e _pr<br>ime | 294478<br>89          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused<br>with ALK exons 20-29) : c.6<br>68-6870:EML4_c.3172+43<br>8:ALKinv   Protein fusion: in<br>frame {EML4-ALK}   | NA                            |

| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>ro<br>le | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost<br>_do<br>mai<br>ns                  | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                              | source_site2<br>_effect_on_f<br>rame |
|----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|-------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|-------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0009<br>339-T02-IM<br>5-50 | P-0009<br>339-T0<br>2-IM5 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294478<br>89               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused<br>with ALK exons 20-29) : c.6<br>68-6870:EML4_c.3172+43<br>8:ALKinv   Protein fusion: in<br>frame (EML4-ALK) | NA                                   |
| EVT-P-0009<br>343-T01-IM<br>5-51 | P-0009<br>343-T0<br>1-IM5 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294465<br>21               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused<br>with ALK exons 20-29) : c.6<br>68-1093:EML4_c.3173-127<br>:ALKinv   Protein fusion: in<br>frame (EML4-ALK) | NA                                   |
| EVT-P-0009<br>353-T01-IM<br>5-52 | P-0009<br>353-T0<br>1-IM5 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294464<br>65               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused<br>with ALK exons 20-29) : c.6<br>67+2407:EML4_c.3173-71:<br>ALKinv   Protein fusion: in<br>frame (EML4-ALK)  | NA                                   |
| EVT-P-0031<br>406-T01-IM<br>6-53 | P-0031<br>406-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294464<br>25               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 with ALK<br>exons 20-29): c.668-2889:<br>EML4_c.3173-31:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}          | NA                                   |
| EVT-P-0032<br>170-T01-IM<br>6-54 | P-0032<br>170-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294476<br>95               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+1822:EML4_c.3172+632<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0010<br>041-T01-IM<br>5-55 | P-0010<br>041-T0<br>1-IM5 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294479<br>42               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                                           |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fuses<br>with ALK exons 20-29) : c.7<br>01-6977:EML4_c.3172+38<br>5:ALKinv   Protein fusion: in<br>frame (EML4-ALK) | NA                                   |
| EVT-P-0008<br>332-T01-IM<br>5-56 | P-0008<br>332-T0<br>1-IM5 |                            | ALK-<br>EML<br>4    | EML<br>4             | unk<br>now<br>n          | five<br>_pr<br>ime      | 294469<br>12               | 20                      | 1058                                | True                           | lost                      | 0.0                              | False                       |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                           | EML4 (NM_019063) - ALK<br>(NM_004304)<br>rearrangement : c.1489+82<br>8:EML4_c.3173-518:ALKd<br>up   Antisense fusion                                                               | NA                                   |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gen_e | fra_me_status        | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                                    | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|------------------|----------------|----------------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|---------------------|---------------------------------|---------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0010<br>177-T01-IM<br>5-57 | P-0010<br>177-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame         | thr_ee<br>_pr<br>ime | 294473<br>58          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv   Protein fusion: in frame (EML4-ALK) | NA                            |
| EVT-P-0007<br>705-T01-IM<br>5-58 | P-0007<br>705-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | unk<br>now<br>n      | thr_ee<br>_pr<br>ime | 294474<br>17          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1594:EML4_c.3172+910:ALKinv   Protein fusion: mid-exon (EML4-ALK)                     | NA                            |
| EVT-P-0010<br>978-T01-IM<br>5-59 | P-0010<br>978-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | out-<br>of-fr<br>ame | thr_ee<br>_pr<br>ime | 294464<br>14          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv   Protein fusion: out of frame (EML4-ALK)       | NA                            |
| EVT-P-0010<br>978-T02-IM<br>5-60 | P-0010<br>978-T0<br>2-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | out-<br>of-fr<br>ame | thr_ee<br>_pr<br>ime | 294464<br>14          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv   Protein fusion: out of frame (EML4-ALK)       | NA                            |
| EVT-P-0011<br>018-T01-IM<br>5-61 | P-0011<br>018-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame         | thr_ee<br>_pr<br>ime | 294483<br>14          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29)   Protein fusion: in frame (EML4-ALK)                                                     | NA                            |
| EVT-P-0011<br>203-T01-IM<br>5-62 | P-0011<br>203-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame         | thr_ee<br>_pr<br>ime | 294471<br>05          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29)   Protein fusion: in frame (EML4-ALK)                                               | NA                            |
| EVT-P-0011<br>203-T02-IM<br>5-63 | P-0011<br>203-T0<br>2-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame         | thr_ee<br>_pr<br>ime | 294470<br>96          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv   Protein fusion: in frame (EML4-ALK)                                                        | NA                            |
| EVT-P-0011<br>203-T03-IM<br>6-64 | P-0011<br>203-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame         | thr_ee<br>_pr<br>ime | 294470<br>55          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv   Protein Fusion: in frame {EML4:ALK}                  | NA                            |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                            | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|------------------|---------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0012<br>679-T01-IM<br>5-65 | P-0012<br>679-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>66          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv   Protein fusion: in frame (EML4-ALK)        | NA                            |
| EVT-P-0012<br>778-T01-IM<br>5-66 | P-0012<br>778-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>42          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKinv   Protein fusion: in frame (EML4-ALK)                                                | NA                            |
| EVT-P-0012<br>778-T02-IM<br>6-67 | P-0012<br>778-T0<br>2-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>42          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                            |
| EVT-P-0007<br>211-T01-IM<br>5-68 | P-0007<br>211-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>89          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165: EML4_c.3172+838:ALKinv   Protein fusion: in frame (EML4-ALK)         | NA                            |
| EVT-P-0007<br>125-T01-IM<br>5-69 | P-0007<br>125-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294465<br>97          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328: EML4_c.3173-203:ALKinv   Protein fusion: mid-exon (EML4-ALK)           | NA                            |
| EVT-P-0006<br>950-T02-IM<br>6-70 | P-0006<br>950-T0<br>2-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294473<br>84          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29): c.1489+2302:EML4_c.3172+943:ALKinv   Protein Fusion: in frame {EML4:ALK}         | NA                            |
| EVT-P-0013<br>202-T01-IM<br>5-71 | P-0013<br>202-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294471<br>78          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565: EML4_c.3173-784:ALKinv   Protein fusion: in frame (EML4-ALK)         | NA                            |

| event_id                         | sampl_e_id                | pa_tie_nt_id | fusio_n_ame      | part_ner_gen_e | fra_me_status | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_f_rame |
|----------------------------------|---------------------------|--------------|------------------|----------------|---------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0006<br>950-T01-IM<br>5-72 | P-0006<br>950-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294473<br>84          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943: ALKInv   Protein fusion: in frame (EML4-ALK)             | NA                            |
| EVT-P-0013<br>453-T01-IM<br>5-73 | P-0013<br>453-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294465<br>97          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKInv   Protein fusion: in frame (EML4-ALK) | NA                            |
| EVT-P-0013<br>864-T01-IM<br>5-74 | P-0013<br>864-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294482<br>09          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118 :ALKInv   Protein fusion: in frame (EML4-ALK)        | NA                            |
| EVT-P-0013<br>869-T01-IM<br>5-75 | P-0013<br>869-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294467<br>12          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318: ALKInv   Protein fusion: in frame (EML4-ALK)        | NA                            |
| EVT-P-0014<br>270-T01-IM<br>6-76 | P-0014<br>270-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294466<br>79          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKInv   Protein fusion: in frame (EML4-ALK)   | NA                            |
| EVT-P-0014<br>310-T01-IM<br>6-77 | P-0014<br>310-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294475<br>44          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20- 29): c.667+6128:EML4_c.3172+783:ALKInv   Protein fusion: in frame (EML4-ALK)            | NA                            |
| EVT-P-0006<br>370-T01-IM<br>5-78 | P-0006<br>370-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294481<br>24          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKInv   Protein fusion: in frame (EML4-ALK)              | NA                            |

| event_id                         | sample_id                 | patient_id | fusion_name      | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0014<br>385-T01-IM<br>6-79 | P-0014<br>385-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294476<br>76       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKinv   Protein fusion: in frame (EML4-ALK)     | NA                           |
| EVT-P-0006<br>313-T01-IM<br>5-80 | P-0006<br>313-T0<br>1-IM5 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294479<br>66       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) -ALK (NM_004304) Fusion ( EML4 exons 1 to 18 fused with exons 20 to 29 of ALK )   Protein fusion: in frame (EML4-ALK)                             | NA                           |
| EVT-P-0030<br>590-T01-IM<br>6-81 | P-0030<br>590-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294477<br>04       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKinv.   Protein Fusion: in frame {EML4:ALK} | NA                           |
| EVT-P-0000<br>474-T01-IM<br>3-82 | P-0000<br>474-T0<br>1-IM3 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294470<br>02       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                           | NA                           |
| EVT-P-0000<br>557-T01-IM<br>3-83 | P-0000<br>557-T0<br>1-IM3 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294465<br>43       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                           | NA                           |
| EVT-P-0000<br>732-T01-IM<br>3-84 | P-0000<br>732-T0<br>1-IM3 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294475<br>37       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                           | NA                           |
| EVT-P-0015<br>197-T01-IM<br>6-85 | P-0015<br>197-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294480<br>09       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20) : c.668-2108:EML4_c.3172+318:ALKinv   Protein fusion: in frame (EML4-ALK)             | NA                           |
| EVT-P-0015<br>434-T02-IM<br>6-86 | P-0015<br>434-T0<br>2-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294481<br>97       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKinv   Protein Fusion: in frame {EML4:ALK}               | NA                           |

| event_id                         | sample_id                 | patient_id | fusion_name      | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                            | source_site2_effect_on_frame |
|----------------------------------|---------------------------|------------|------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0005<br>321-T01-IM<br>5-87 | P-0005<br>321-T0<br>1-IM5 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294482<br>50       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exon 13 fused with<br>ALK exon 20): c.2068+156<br>:EML4_c.3173-693:ALKinv<br>  Protein fusion: in frame<br>(EML4-ALK)    | NA                           |
| EVT-P-0029<br>971-T01-IM<br>6-88 | P-0029<br>971-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294478<br>07       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-119:EML4_c.3172+520:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                           |
| EVT-P-0015<br>921-T03-IM<br>6-89 | P-0015<br>921-T0<br>3-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294468<br>09       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused<br>with ALK exons 20-29): c.6<br>67+516:EML4_c.3173-415:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                           |
| EVT-P-0015<br>961-T01-IM<br>6-90 | P-0015<br>961-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294465<br>26       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304 ) fusion<br>(EML4 exon 6fused to ALK<br>exon 20): c.668-6666:EML<br>4_c.3173-132:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}          | NA                           |
| EVT-P-0015<br>961-T02-IM<br>6-91 | P-0015<br>961-T0<br>2-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294465<br>26       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused to<br>ALK exons 20-29): c.668-6<br>666:EML4_c.3173-132:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                           |
| EVT-P-0015<br>961-T03-IM<br>6-92 | P-0015<br>961-T0<br>3-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294465<br>26       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused to<br>ALK exons 20-29): c.668-6<br>666:EML4_c.3173-132:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                           |
| EVT-P-0031<br>827-T01-IM<br>6-93 | P-0031<br>827-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294467<br>00       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-673:EML4_c.3173-306:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                  | NA                           |
| EVT-P-0027<br>119-T01-IM<br>6-94 | P-0027<br>119-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294467<br>11       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-12 fused to<br>ALK exons 20-29): c.1490-<br>837:EML4_c.3173-317:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                           |

| event_id                          | sampl_e_id                | pa_tie_n_t_id | fusio_n_name     | part_ner_gene | fra_me_status | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                           | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|---------------|------------------|---------------|---------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0029<br>750-T01-IM<br>6-95  | P-0029<br>750-T0<br>1-IM6 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294466<br>60          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv   Protein Fusion: in frame {EML4:ALK}   | NA                            |
| EVT-P-0029<br>550-T01-IM<br>6-96  | P-0029<br>550-T0<br>1-IM6 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294478<br>63          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                            |
| EVT-P-0016<br>280-T01-IM<br>6-97  | P-0016<br>280-T0<br>1-IM6 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294479<br>05          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv   Protein Fusion: in frame {EML4:ALK}                                       | NA                            |
| EVT-P-0016<br>580-T01-IM<br>6-98  | P-0016<br>580-T0<br>1-IM6 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294475<br>79          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv   Protein Fusion: in frame {EML4:ALK}   | NA                            |
| EVT-P-0005<br>305-T01-IM<br>5-99  | P-0005<br>305-T0<br>1-IM5 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294477<br>64          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALKinv   Protein fusion: in frame {EML4:ALK}         | NA                            |
| EVT-P-0016<br>960-T01-IM<br>6-100 | P-0016<br>960-T0<br>1-IM6 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294478<br>38          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv   Protein Fusion: in frame {EML4:ALK}     | NA                            |
| EVT-P-0017<br>592-T01-IM<br>5-101 | P-0017<br>592-T0<br>1-IM5 |               | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr<br>ee<br>_pr<br>ime | 294475<br>48          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                                        | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|---------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0029<br>400-T01-IM<br>6-102 | P-0029<br>400-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294476<br>68          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 20-29): c.1<br>315+641:EML4_c.3172+65<br>9:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}        | NA                            |
| EVT-P-0004<br>218-T01-IM<br>5-103 | P-0004<br>218-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294470<br>87          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exon 20 fused with<br>ALK exon 20) : c.2068+156<br>:EML4_c.3173-693:ALKinv<br>  Protein fusion: in frame<br>(EML4-ALK)               | NA                            |
| EVT-P-0017<br>978-T01-IM<br>6-104 | P-0017<br>978-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294466<br>09          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused to<br>ALK exons 20-29): c.667+7<br>306:EML4_c.3173-215:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}              | NA                            |
| EVT-P-0017<br>978-T02-IM<br>6-105 | P-0017<br>978-T0<br>2-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294466<br>09          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused to<br>ALK exons 20-29): c.667+7<br>306:EML4_c.3173-215:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}              | NA                            |
| EVT-P-0017<br>978-T03-IM<br>6-106 | P-0017<br>978-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294466<br>09          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 fused to<br>ALK exons 20-29): c.667+7<br>306:EML4_c.3173-215:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}              | NA                            |
| EVT-P-0018<br>242-T01-IM<br>6-107 | P-0018<br>242-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294480<br>07          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 fused<br>in-frame with ALK exons<br>20-29): c.668-7560:EML4_<br>c.3172+320:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK} | NA                            |
| EVT-P-0018<br>280-T01-IM<br>6-108 | P-0018<br>280-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294471<br>87          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exon 6 fused with<br>ALK exon 20): c.494-6792:<br>EML4_c.3173-793:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}                 | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_j_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                                                                                                  | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|----------------|------------------|---------------|-----------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|---------------------|---------------------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0003<br>913-T01-IM<br>3-109 | P-0003<br>913-T0<br>1-IM3 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294472<br>11          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | NA   Protein fusion: in<br>frame (EML4-ALK)                                                                                                                                                                                             | NA                            |
| EVT-P-0029<br>174-T01-IM<br>6-110 | P-0029<br>174-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294466<br>56          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-19 fused<br>with ALK exons 20-29): c.2<br>242+168:EML4_c.3173-26<br>2:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}                                                  | NA                            |
| EVT-P-0018<br>969-T01-IM<br>6-112 | P-0018<br>969-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294484<br>16          | 19               | 1028                         | False                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-760:EML4_c.3083:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                                                                            | NA                            |
| EVT-P-0019<br>491-T01-IM<br>6-113 | P-0019<br>491-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294465<br>96          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-12 with<br>ALK exons 20-29): c.1490-<br>486:EML4_c.3173-202:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                                           | NA                            |
| EVT-P-0019<br>513-T01-IM<br>6-114 | P-0019<br>513-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294484<br>03          | 19               | 1032                         | False                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 19-29): c.1<br>489+2278:EML4_c.3096:A<br>LKinv   Protein Fusion:<br>mid-exon {EML4:ALK}                                                        | NA                            |
| EVT-P-0019<br>533-T01-IM<br>6-115 | P-0019<br>533-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294464<br>94          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-6 with ALK<br>exons 20-29): c.667+6837:<br>EML4_c.3173-100:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}                                                             | NA                            |
| EVT-P-0009<br>102-T01-IM<br>5-116 | P-0009<br>102-T0<br>1-IM5 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294465<br>13          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-12 fused in<br>frame with ALK exons<br>20-29, which include the<br>ALK kinase domain): c.149<br>0-244:EML4_c.3173-119:A<br>LKinv   Protein fusion: in<br>frame (EML4-ALK) | NA                            |
| EVT-P-0028<br>901-T01-IM<br>6-117 | P-0028<br>901-T0<br>1-IM6 |                | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294466<br>07          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 20-29): c.1<br>316-844:EML4_c.3173-213<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                  | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gen_e | fra_me_status | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_nicated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|----------------|---------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|-----------------------|---------------------------------|---------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0019<br>786-T01-IM<br>6-118 | P-0019<br>786-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294475<br>75          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKinv   Protein Fusion: in frame {EML4:ALK}  | NA                            |
| EVT-P-0019<br>972-T01-IM<br>6-119 | P-0019<br>972-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294481<br>01          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKinv   Protein Fusion: in frame {EML4:ALK}          | NA                            |
| EVT-P-0020<br>124-T01-IM<br>6-120 | P-0020<br>124-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294479<br>55          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                            |
| EVT-P-0020<br>138-T01-IM<br>6-121 | P-0020<br>138-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294479<br>07          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-3016:EML4_c.3172+420:ALKinv   Protein Fusion: in frame {EML4:ALK}            | NA                            |
| EVT-P-0003<br>383-T01-IM<br>5-122 | P-0003<br>383-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294475<br>72          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                                 | NA                            |
| EVT-P-0020<br>408-T01-IM<br>6-123 | P-0020<br>408-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294467<br>79          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:EML4_c.3173-385:ALKinv   Protein Fusion: in frame {EML4:ALK}           | NA                            |
| EVT-P-0003<br>323-T01-IM<br>5-124 | P-0003<br>323-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294475<br>30          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                                 | NA                            |
| EVT-P-0003<br>297-T01-IM<br>5-125 | P-0003<br>297-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame  | thr_ee<br>_pr<br>ime | 294482<br>59          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                                 | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intronic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains               | lost_domains | disrupt_ed_domains | source_annotation_text                                                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|-----------------|----------------------|-----------------------|------------------|------------------------------|-------------------------|----------------|----------------------------|---------------------|---------------------------------|--------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0020<br>584-T01-IM<br>6-126 | P-0020<br>584-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294470<br>48          | 20               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-2 fused to<br>ALK exon 20-29) : c.209-48<br>46:EML4_c.3173-654:ALKi<br>nv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                           |
| EVT-P-0020<br>780-T01-IM<br>6-127 | P-0020<br>780-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294470<br>84          | 20               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1 - 13 fused<br>to ALK exons 20 - 29) : c.1<br>489+1278:EML4_c.3173-6<br>90:ALKinv   Protein Fusion:<br>in frame {EML4:ALK} | NA                           |
| EVT-P-0020<br>825-T03-IM<br>6-128 | P-0020<br>825-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294467<br>67          | 20               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-2 fused to<br>ALK exons 20-29):c.208+4<br>890:EML4_c.3173-373:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}         | NA                           |
| EVT-P-0021<br>257-T01-IM<br>6-129 | P-0021<br>257-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294477<br>77          | 19               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exons 1-17 fused<br>with ALK exons 20-29): c.1<br>883-596:EML4_c.3172+55<br>0:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}  | NA                           |
| EVT-P-0003<br>206-T01-IM<br>5-130 | P-0003<br>206-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294483<br>65          | 19               | 1045                         | False                   | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | NA   Protein fusion:<br>mid-exon (EML4-ALK)                                                                                                                                             | NA                           |
| EVT-P-0021<br>495-T01-IM<br>6-131 | P-0021<br>495-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294483<br>13          | 19               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exons 1-5 fused<br>with ALK exons 20-29): c.4<br>93+2355:EML4_c.3172+14<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}    | NA                           |
| EVT-P-0002<br>613-T01-IM<br>3-132 | P-0002<br>613-T0<br>1-IM3 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294471<br>99          | 20               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | NA   Protein fusion: in<br>frame (EML4-ALK)                                                                                                                                             | NA                           |
| EVT-P-0002<br>408-T02-IM<br>5-133 | P-0002<br>408-T0<br>2-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294480<br>53          | 19               | 1058                         | True                    | retain<br>ed   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |              |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-18 fuses<br>with ALK exons 20-29) : c.2<br>057-115:EML4_c.3172+27<br>4:ALKinv   Protein fusion: in<br>frame (EML4-ALK)    | NA                           |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>ro<br>le | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost_<br>do<br>mains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                  | source_site2<br>_effect_on_f<br>rame |
|-----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|-------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|----------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0021<br>693-T01-IM<br>6-134 | P-0021<br>693-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294470<br>02               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_001145076) -<br>ALK (NM_004304) Fusion<br>(EML4 exons 1-14 fused<br>with ALK exons 20-29): c.1<br>316-1509:EML4_c.3173-60<br>8:ALKinv   Protein Fusion:<br>in frame {EML4:ALK} | NA                                   |
| EVT-P-0022<br>101-T01-IM<br>6-135 | P-0022<br>101-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294477<br>59               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) Fusion<br>(EML4 exon18 fused with<br>ALK exon 20): c.2057-510:<br>EML4_c.2057-510:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}              | NA                                   |
| EVT-P-0022<br>138-T01-IM<br>6-136 | P-0022<br>138-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | unk<br>now<br>n          | thre<br>_pr<br>ime      | 294465<br>84               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 fused<br>with ALK exons 20-29): c.2<br>333:EML4_c.3173-190:AL<br>Kinv   Protein Fusion:<br>mid-exon {EML4:ALK}         | NA                                   |
| EVT-P-0036<br>247-T01-IM<br>6-137 | P-0036<br>247-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294478<br>57               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+107:EML4_c.3172+470:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                        | NA                                   |
| EVT-P-0036<br>106-T01-IM<br>6-138 | P-0036<br>106-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294469<br>30               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) -ALK<br>(NM_004304) fusion<br>(EML4 exons 1-13 fused to<br>ALK exons 20-29): c.1490-<br>2055:EML4_c.3173-536:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}       | NA                                   |
| EVT-P-0028<br>745-T01-IM<br>6-139 | P-0028<br>745-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294475<br>09               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 20-29): c.1<br>523-2563:EML4_c.3172:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                                   |
| EVT-P-0022<br>398-T01-IM<br>6-140 | P-0022<br>398-T0<br>1-IM6 |                            | ALK-<br>EML<br>4    | EML<br>4             | in-fr<br>ame             | thre<br>_pr<br>ime      | 294480<br>35               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-5 fused<br>with ALK exons 20-29): c.4<br>94-7853:EML4_c.3172+29<br>2:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}   | NA                                   |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status | tar_get_role       | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains                             | disrupt_ed_domains | source_annotation_text                                                                                                                                                                                               | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|---------------|--------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|-------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0002<br>266-T01-IM<br>3-141 | P-0002<br>266-T0<br>1-IM3 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | five<br>_pr<br>ime | 294472<br>88          | 20               | 1058                         | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                    | ALK (NM_004304) and<br>EML4 (NM_019063)<br>reciprocal inversion<br>(13026566bp): c.3173-889:<br>ALK_c.208+1022:EML4inv<br>  Protein fusion: in frame<br>(ALK-EML4)                                                   | NA                            |
| EVT-P-0002<br>266-T01-IM<br>3-142 | P-0002<br>266-T0<br>1-IM3 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294472<br>83          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | NA   Protein fusion: in<br>frame (EML4-ALK)                                                                                                                                                                          | NA                            |
| EVT-P-0028<br>541-T03-IM<br>6-143 | P-0028<br>541-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294469<br>22          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion:<br>c.208+4631:EML4_c.3173-<br>528:ALKinv   Protein<br>Fusion: in frame<br>{EML4:ALK}                                                                                | NA                            |
| EVT-P-0022<br>926-T01-IM<br>6-144 | P-0022<br>926-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294465<br>58          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-5 fused<br>with ALK exons 20-29): c.4<br>93+983:EML4_c.3173-164:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                 | NA                            |
| EVT-P-0035<br>121-T01-IM<br>6-145 | P-0035<br>121-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294473<br>83          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-4857:EML4_c.3172+944:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                                                     | NA                            |
| EVT-P-0002<br>217-T02-IM<br>3-146 | P-0002<br>217-T0<br>2-IM3 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294465<br>94          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | NA   Protein fusion: in<br>frame (EML4-ALK)                                                                                                                                                                          | NA                            |
| EVT-P-0023<br>368-T01-IM<br>6-147 | P-0023<br>368-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294467<br>13          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 fused<br>with ALK exons 20-29<br>including the kinase<br>domain) : c.667+7004:EML<br>4_c.3173-319:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK} | NA                            |
| EVT-P-0002<br>024-T01-IM<br>3-148 | P-0002<br>024-T0<br>1-IM3 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame  | thre<br>_pr<br>ime | 294471<br>33          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) inversion: c.<br>2242+31:EML4_c.3173-73<br>9:ALKinv   Protein fusion: in<br>frame (EML4-ALK)                                                                                   | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name      | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                            | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0023<br>962-T01-IM<br>6-149 | P-0023<br>962-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294488<br>20       | 19              | 1023                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:EML4_c.3068-389:ALKinv   Protein Fusion: in frame {EML4:ALK}    | NA                           |
| EVT-P-0024<br>136-T01-IM<br>6-150 | P-0024<br>136-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294482<br>66       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498:EML4_c.3172+61:ALKinv   Protein Fusion: in frame {EML4:ALK}      | NA                           |
| EVT-P-0001<br>811-T01-IM<br>3-151 | P-0001<br>811-T0<br>1-IM3 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294482<br>05       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EML4_c.3172+122:ALKinv   Protein fusion: in frame (EML4-ALK)                                         | NA                           |
| EVT-P-0028<br>541-T02-IM<br>6-152 | P-0028<br>541-T0<br>2-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294469<br>22       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                           |
| EVT-P-0024<br>907-T01-IM<br>6-153 | P-0024<br>907-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294468<br>77       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209-990:EML4_c.3173-483:ALKinv   Protein Fusion: in frame {EML4:ALK}      | NA                           |
| EVT-P-0034<br>569-T02-IM<br>6-154 | P-0034<br>569-T0<br>2-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294465<br>14       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKinv   Protein Fusion: in frame {EML4:ALK}     | NA                           |
| EVT-P-0034<br>450-T01-IM<br>6-155 | P-0034<br>450-T0<br>1-IM6 |            | ALK-<br>EML<br>4 | EML<br>4     | unknown      | three_prime | 294463<br>54       | 20              | 1071                        | False                  | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKinv   Protein Fusion: mid-exon {EML4:ALK}             | NA                           |
| EVT-P-0001<br>532-T01-IM<br>3-156 | P-0001<br>532-T0<br>1-IM3 |            | ALK-<br>EML<br>4 | EML<br>4     | in-frame     | three_prime | 294475<br>53       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | NA   Protein fusion: in frame (EML4-ALK)                                                                                                                          | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_ame      | part_ner_gen_e | fra_me_status   | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains                             | disrupt_ed_domains | source_annotation_text                                                                                                                                                                       | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|----------------|-----------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|-------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0028<br>541-T01-IM<br>6-157 | P-0028<br>541-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294469<br>22          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-2 fused<br>with ALK exons 20-29): c.2<br>08+4631:EML4_c.3173-52<br>8:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}        | NA                            |
| EVT-P-0026<br>407-T01-IM<br>6-158 | P-0026<br>407-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294463<br>92          | 20               | 1059                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-18 fused<br>with ALK exons 20-29): c.1<br>981-2450:EML4_c.3175:AL<br>Kinv   Protein Fusion:<br>mid-exon {EML4:ALK}          | NA                            |
| EVT-P-0026<br>650-T02-IM<br>6-159 | P-0026<br>650-T0<br>2-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame    | five<br>_pr<br>ime   | 294474<br>51          | 19               | 1058                         | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 fuses<br>with ALK exons 20-29) : c.6<br>67+7374:EML4_c.3172+87<br>6:ALKinv   Protein Fusion:<br>in frame {ALK:EML4}          | NA                            |
| EVT-P-0026<br>657-T01-IM<br>6-160 | P-0026<br>657-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294469<br>52          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 fuses<br>with ALK exons 20-29) : c.2<br>312:EML4_c.3173-558:AL<br>Kinv   Protein Fusion:<br>mid-exon {EML4:ALK}             | NA                            |
| EVT-P-0026<br>657-T03-IM<br>6-161 | P-0026<br>657-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294469<br>52          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-5 fused to<br>ALK exons 20-29): c.2312:<br>EML4_c.3173-558:ALKinv  <br>Protein Fusion: mid-exon<br>{EML4:ALK}                  | NA                            |
| EVT-P-0015<br>152-T02-IM<br>6-162 | P-0015<br>152-T0<br>2-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294470<br>79          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 fused<br>in-frame to ALK exons<br>20-29): c.2242+127:EML4_<br>c.3173-685:ALKinv  <br>Protein fusion: in frame<br>(EML4-ALK) | NA                            |
| EVT-P-0032<br>862-T01-IM<br>6-163 | P-0032<br>862-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4       | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294474<br>88          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | EML4 (NM_019063) - ALK<br>(NM_004304)<br>Fusion(EML4 exon 11<br>fused to ALK exon20 ): c.1<br>489+671:EML4_c.3172+83<br>9:ALK   Protein Fusion: in<br>frame {EML4:ALK}                       | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                                                 | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|-----------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0027<br>119-T03-IM<br>6-164 | P-0027<br>119-T0<br>3-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294467<br>11          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 20-29): c.1<br>316-837:EML4_c.3173-317<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                            |
| EVT-P-0032<br>683-T01-IM<br>6-165 | P-0032<br>683-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294477<br>63          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2122:EML4_c.3172+564:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                      | NA                            |
| EVT-P-0028<br>339-T01-IM<br>6-166 | P-0028<br>339-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294476<br>80          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-2 fused<br>with ALK exons 20-29): c.2<br>08+1245:EML4_c.3172+64<br>7:ALKinv   Protein Fusion:<br>in frame {EML4:ALK}  | NA                            |
| EVT-P-0032<br>518-T01-IM<br>6-167 | P-0032<br>518-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | unk<br>now<br>n | thr_ee<br>_pr<br>ime | 294483<br>59          | 19               | 1047                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion:<br>c.1315+2607:EML4_c.314<br>0:ALKinv   Protein Fusion:<br>mid-exon {EML4:ALK}                                                        | NA                            |
| EVT-P-0027<br>853-T01-IM<br>6-168 | P-0027<br>853-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294477<br>34          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 fused to<br>ALK exons 20-29 ): c.2243-<br>157:EML4_c.3172+593:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                            |
| EVT-P-0015<br>110-T01-IM<br>6-169 | P-0015<br>110-T0<br>1-IM6 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294465<br>18          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) Fusion<br>(EML4 exons 1-5 with ALK<br>exons 20-29) : c.668-1726:<br>EML4_c.3173-124:ALKinv  <br>Protein fusion: in frame<br>(EML4-ALK)           | NA                            |
| EVT-P-0003<br>446-T01-IM<br>5-170 | P-0003<br>446-T0<br>1-IM5 |              | ALK-<br>EML<br>4 | EML<br>4      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294465<br>34          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | NA   Protein fusion: in<br>frame (EML4-ALK)                                                                                                                                            | NA                            |
| EVT-P-0004<br>516-T02-IM<br>5-171 | P-0004<br>516-T0<br>2-IM5 |              | ALK-<br>GCC<br>2 | GC<br>C2      | in-fr<br>ame    | thr_ee<br>_pr<br>ime | 294465<br>26          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | GCC2 (NM_181453) - ALK<br>(NM_004304)<br>rearrangement : c.4448-12<br>30:GCC2_c.3173-132:ALKi<br>nv   Protein fusion: in frame<br>(GCC2-ALK)                                           | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                         | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0035<br>712-T01-IM<br>6-172 | P-0035<br>712-T0<br>1-IM6 |            | ALK-H6PD    | H6PD         | unknown      | five_prime  | 29462508           | 13              | 785                         | True                   | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | ALK (NM_004304) rearrangement: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr2:g.29462508)   Antisense Fusion                                                        | NA                           |
| EVT-P-0018<br>541-T01-IM<br>6-174 | P-0018<br>541-T0<br>1-IM6 |            | ALK-LRP1B   | LRP1B        | in-frame     | five_prime  | 29450289           | 17              | 972                         | True                   | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | ALK (NM_004304) - LRP1B (NM_018557) Rearrangement : c.2914+151:ALK_c.344-85984:LRP1Bdup   Protein Fusion: in frame {ALK:LRP1B}                                 | NA                           |
| EVT-P-0027<br>092-T01-IM<br>6-176 | P-0027<br>092-T0<br>1-IM6 |            | ALK-PLEKHH2 | PLEKHH2      | in-frame     | three_prime | 29446506           | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545:PLEKHH2_c.3173-112:ALKinv   Protein Fusion: in frame {PLEKHH2:ALK}                              | NA                           |
| EVT-P-0027<br>363-T02-IM<br>6-177 | P-0027<br>363-T0<br>2-IM6 |            | ALK-RANBP2  | RANBP2       | in-frame     | three_prime | 29447364           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv   Protein Fusion: in frame {RANBP2:ALK}                               | NA                           |
| EVT-P-0068<br>865-T02-IM<br>7-178 | P-0068<br>865-T0<br>2-IM7 |            | ALK-RANBP2  | RANBP2       | in-frame     | five_prime  | 29447373           | 19              | 1058                        | True                   | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2inv   Protein Fusion: in frame {ALK:RANBP2}                               | NA                           |
| EVT-P-0008<br>843-T01-IM<br>5-179 | P-0008<br>843-T0<br>1-IM5 |            | ALK-RANBP2  | RANBP2       | in-frame     | three_prime | 29446488           | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv   Protein fusion: in frame (RANBP2-ALK)                                | NA                           |
| EVT-P-0015<br>699-T01-IM<br>6-181 | P-0015<br>699-T0<br>1-IM6 |            | ALK-SPTBN1  | SPTBN1       | in-frame     | three_prime | 29447437           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028: SPTBN1_c.3172+890:ALKinv   Protein fusion: in frame (SPTBN1-ALK) | NA                           |
| EVT-P-0016<br>193-T05-IM<br>6-182 | P-0016<br>193-T0<br>5-IM6 |            | ALK-STRN    | STRN         | unknown      | three_prime | 29448326           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel   Protein Fusion: mid-exon {STRN:ALK}   | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name       | part_ner_gene | fra_me_status        | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains                             | disrupt_ed_domains | source_annotation_text                                                                                                                             | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|--------------------|---------------|----------------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|-------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0016<br>193-T02-IM<br>6-183 | P-0016<br>193-T0<br>2-IM6 |              | ALK-STR<br>N       | STR<br>N      | unk<br>now<br>n      | thr<br>ee<br>_pr<br>ime | 294483<br>26          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel   Protein Fusion: mid-exon {STRN:ALK}                                         | NA                            |
| EVT-P-0016<br>193-T01-IM<br>6-184 | P-0016<br>193-T0<br>1-IM6 |              | ALK-STR<br>N       | STR<br>N      | unk<br>now<br>n      | thr<br>ee<br>_pr<br>ime | 294483<br>26          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel   Protein Fusion: mid-exon {STRN:ALK}                                         | NA                            |
| EVT-P-0041<br>438-T01-IM<br>6-185 | P-0041<br>438-T0<br>1-IM6 |              | ALK-TTC<br>27      | TTC<br>27     | out-<br>of-fr<br>ame | five<br>_pr<br>ime      | 294478<br>17          | 19               | 1058                         | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                    | ALK (NM_004304) - TTC27 (NM_017735) rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv   Protein Fusion: out of frame {ALK:TTC27}                   | NA                            |
| EVT-P-0006<br>617-T01-IM<br>5-187 | P-0006<br>617-T0<br>1-IM5 |              | ASA<br>P2-A<br>LK  | ASA<br>P2     | unk<br>now<br>n      | five<br>_pr<br>ime      | 301427<br>53          | 1                | 223                          | True                     | lost           | 0.0                        | False                |                                 | Prot<br>ein k<br>inas<br>e do<br>mai<br>n |                    | ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel   Antisense fusion                                    | NA                            |
| EVT-P-0062<br>994-T02-IM<br>7-188 | P-0062<br>994-T0<br>2-IM7 |              | CLT<br>C-AL<br>K   | CLT<br>C      | in-fr<br>ame         | thr<br>ee<br>_pr<br>ime | 294496<br>24          | 18               | 1023                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29449624::chr17:g.57768350)   Protein Fusion: in frame {CLTC:ALK}           | NA                            |
| EVT-P-0051<br>864-T01-IM<br>6-189 | P-0051<br>864-T0<br>1-IM6 |              | CLT<br>C-AL<br>K   | CLT<br>C      | in-fr<br>ame         | thr<br>ee<br>_pr<br>ime | 294488<br>39          | 19               | 1023                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29448839::chr17:g.57770376)   Protein Fusion: in frame {CLTC:ALK}           | NA                            |
| EVT-P-0025<br>388-T01-IM<br>6-190 | P-0025<br>388-T0<br>1-IM6 |              | CTN<br>NA1-<br>ALK | CTN<br>NA1    | in-fr<br>ame         | thr<br>ee<br>_pr<br>ime | 294464<br>65          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | CTNNA1 (NM_001903) - ALK (NM_004304) rearrangement: t(2;5)(p23.2;q31.2)(chr2:g.29446465::chr5:g.138259019)   Protein Fusion: in frame {CTNNA1:ALK} | NA                            |
| EVT-P-0051<br>068-T01-IM<br>6-191 | P-0051<br>068-T0<br>1-IM6 |              | DIAP<br>H2-A<br>LK | DIA<br>PH2    | in-fr<br>ame         | thr<br>ee<br>_pr<br>ime | 294475<br>66          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |                                           |                    | DIAPH2 (NM_006729) - ALK (NM_004304) fusion: t(2;X)(p23.2;q21.33)(chr2:g.29447566::chrX:g.96713620)   Protein Fusion: in frame {DIAPH2:ALK}        | NA                            |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>role      | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost_<br>do<br>mains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                             | source_site2<br>_effect_on_f<br>rame |
|-----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|--------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|----------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0005<br>759-T02-IM<br>6-192 | P-0005<br>759-T0<br>2-IM6 |                            | EMIL<br>IN1-<br>ALK | EMI<br>LIN1          | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294507<br>41               | 17                      | 939                                 | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EMILIN1 (NM_007046) -<br>ALK (NM_004304) fusion:<br>c.2575+44:EMILIN1_c.281<br>6-203:ALKinv   Protein<br>Fusion: in frame<br>{EMILIN1:ALK}                                         | NA                                   |
| EVT-P-0046<br>083-T01-IM<br>6-193 | P-0046<br>083-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294466<br>65               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+986:EML4_c.3173-271:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0049<br>375-T01-IM<br>6-194 | P-0049<br>375-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294467<br>32               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+1166:EML4_c.3173-338:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0015<br>646-T03-IM<br>6-195 | P-0015<br>646-T0<br>3-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294480<br>28               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.209<br>-3909:EML4_c.3172+299:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0014<br>385-T03-IM<br>6-196 | P-0014<br>385-T0<br>3-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294476<br>76               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+1890:EML4_c.3172+651<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                  | NA                                   |
| EVT-P-0027<br>119-T04-IM<br>7-197 | P-0027<br>119-T0<br>4-IM7 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294467<br>11               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-837:EML4_c.3173-317:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0057<br>391-T02-IM<br>6-198 | P-0057<br>391-T0<br>2-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294470<br>92               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-3018:EML4_c.3173-698:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0034<br>334-T01-IM<br>6-199 | P-0034<br>334-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294465<br>70               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4909:EML4_c.3173-176:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                                   | NA                                   |
| EVT-P-0025<br>095-T01-IM<br>6-200 | P-0025<br>095-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294473<br>21               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 fused to<br>ALK exons 20 -29): c.2243-<br>278:EML4_c.3173-927:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                                   |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gen_e | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_truncated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                                                  | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|----------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|---------------------|---------------------------------|---------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0025<br>040-T01-IM<br>6-201 | P-0025<br>040-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294471<br>52          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exon 12 fused to<br>ALK exon 20) : c.1316-194<br>0:EML4_c.3173-758:ALKIn<br>v   Protein Fusion: in frame<br>{EML4:ALK}         | NA                            |
| EVT-P-0007<br>705-T02-IM<br>6-202 | P-0007<br>705-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294474<br>17          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.159<br>4:EML4_c.3172+910:ALKIn<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                            | NA                            |
| EVT-P-0023<br>375-T02-IM<br>7-203 | P-0023<br>375-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294478<br>54          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2728:EML4_c.3172+473:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                       | NA                            |
| EVT-P-0035<br>085-T03-IM<br>6-204 | P-0035<br>085-T0<br>3-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>79          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+7361:EML4_c.3173-685:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                        | NA                            |
| EVT-P-0001<br>766-T04-IM<br>6-205 | P-0001<br>766-T0<br>4-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294487<br>67          | 19               | 1023                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.208<br>+3962:EML4_c.3068-336:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                        | NA                            |
| EVT-P-0068<br>467-T02-IM<br>7-206 | P-0068<br>467-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294482<br>95          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+1566:EML4_c.3172+32:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                        | NA                            |
| EVT-P-0022<br>184-T04-IM<br>6-207 | P-0022<br>184-T0<br>4-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>35          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+1386:EML4_c.3173-641:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                       | NA                            |
| EVT-P-0036<br>106-T03-IM<br>7-208 | P-0036<br>106-T0<br>3-IM7 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294469<br>30          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2055:EML4_c.3173-536:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                                       | NA                            |
| EVT-P-0022<br>184-T01-IM<br>6-209 | P-0022<br>184-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>35          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False               | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_001145076) -<br>ALK (NM_004304) fusion<br>(EML4 exons 1-12 fused<br>with ALK exons 20-29): c.1<br>315+1386:EML4_c.3173-6<br>41:ALKInv   Protein Fusion:<br>in frame {EML4:ALK} | NA                            |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>ro<br>le  | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost_<br>do<br>mains | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                     | source_site2<br>_effect_on_f<br>rame |
|-----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|--------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|----------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0037<br>404-T01-IM<br>6-210 | P-0037<br>404-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294478<br>72               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4963:EML4_c.3172+455:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0037<br>491-T01-IM<br>6-211 | P-0037<br>491-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294467<br>86               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) Fusion<br>(EML4 exon12 fused with<br>ALK exon20): c.1490-1120:<br>EML4_c.3173-392:ALKInv  <br>Protein Fusion: in frame<br>{EML4:ALK} | NA                                   |
| EVT-P-0037<br>668-T01-IM<br>6-212 | P-0037<br>668-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294476<br>31               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2742:EML4_c.3172+696:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                          | NA                                   |
| EVT-P-0037<br>925-T01-IM<br>6-213 | P-0037<br>925-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294465<br>22               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4634:EML4_c.3173-128:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0037<br>982-T01-IM<br>6-214 | P-0037<br>982-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294464<br>79               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-138:EML4_c.3173-85:AL<br>KInv   Protein Fusion: in<br>frame {EML4:ALK}                                            | NA                                   |
| EVT-P-0038<br>201-T03-IM<br>6-215 | P-0038<br>201-T0<br>3-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294477<br>48               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+989:EML4_c.3172+579:<br>ALKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0038<br>772-T01-IM<br>6-216 | P-0038<br>772-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294466<br>87               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.234<br>1+147:EML4_c.3173-293:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0043<br>860-T02-IM<br>6-217 | P-0043<br>860-T0<br>2-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294469<br>31               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-980:EML4_c.3173-537:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0067<br>093-T02-IM<br>7-218 | P-0067<br>093-T0<br>2-IM7 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thr<br>ee_<br>_pr<br>ime | 294473<br>03               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                      |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.208<br>+1508:EML4_c.3173-909:A<br>LKInv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |

| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>ro<br>le  | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_tru<br>ncated | retain<br>ed_do<br>mains        | lost_<br>do<br>mai<br>ns | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                    | source_site2<br>_effect_on_f<br>rame |
|-----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|--------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|--------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0017<br>592-T04-IM<br>6-219 | P-0017<br>592-T0<br>4-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294475<br>48               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-5963:EML4_c.3172+779:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                          | NA                                   |
| EVT-P-0040<br>261-T01-IM<br>6-220 | P-0040<br>261-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294471<br>03               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-1340:EML4_c.3173-709:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                         | NA                                   |
| EVT-P-0015<br>961-T07-IM<br>6-221 | P-0015<br>961-T0<br>7-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294465<br>26               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-6666:EML4_c.3173-132:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                          | NA                                   |
| EVT-P-0040<br>553-T01-IM<br>6-222 | P-0040<br>553-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294470<br>55               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+30:EML4_c.3173-661:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                           | NA                                   |
| EVT-P-0040<br>574-T01-IM<br>6-223 | P-0040<br>574-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294477<br>38               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exon5 fused with<br>ALK exon20) : c.668-1589:<br>EML4_c.3172+589:ALKinv<br>  Protein Fusion: in frame<br>{EML4:ALK} | NA                                   |
| EVT-P-0040<br>576-T01-IM<br>6-224 | P-0040<br>576-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294473<br>21               | 20                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exon5 fused with<br>ALK exon20) : c.668-821:E<br>ML4_c.3173-927:ALKinv  <br>Protein Fusion: in frame<br>{EML4:ALK}  | NA                                   |
| EVT-P-0040<br>600-T01-IM<br>6-225 | P-0040<br>600-T0<br>1-IM6 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294476<br>88               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+1795:EML4_c.3172+639:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                          | NA                                   |
| EVT-P-0032<br>683-T02-IM<br>7-226 | P-0032<br>683-T0<br>2-IM7 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294477<br>63               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2122:EML4_c.3172+564:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                         | NA                                   |
| EVT-P-0014<br>385-T06-IM<br>7-227 | P-0014<br>385-T0<br>6-IM7 |                            | EML<br>4-AL<br>K    | EML<br>4             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294476<br>76               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+1890:EML4_c.3172+651<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                         | NA                                   |

| event_id                          | sample_id                 | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains            | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                  | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0067<br>069-T01-IM<br>7-228 | P-0067<br>069-T0<br>1-IM7 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294481<br>69       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKinv   Protein Fusion: in frame {EML4:ALK}                                     | NA                           |
| EVT-P-0014<br>385-T02-IM<br>6-229 | P-0014<br>385-T0<br>2-IM6 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294476<br>76       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv   Protein Fusion: in frame {EML4:ALK}                                     | NA                           |
| EVT-P-0041<br>967-T01-IM<br>6-230 | P-0041<br>967-T0<br>1-IM6 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294481<br>47       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv   Protein Fusion: in frame {EML4:ALK}                                      | NA                           |
| EVT-P-0041<br>967-T02-IM<br>7-231 | P-0041<br>967-T0<br>2-IM7 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294481<br>47       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv   Protein Fusion: in frame {EML4:ALK}                                      | NA                           |
| EVT-P-0067<br>048-T02-IM<br>7-232 | P-0067<br>048-T0<br>2-IM7 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294473<br>91       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2216:EML4_c.3172+936:ALKinv   Protein Fusion: in frame {EML4:ALK}                                      | NA                           |
| EVT-P-0066<br>568-T02-IM<br>7-233 | P-0066<br>568-T0<br>2-IM7 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294480<br>72       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+3428:EML4_c.3172+255:ALKinv   Protein Fusion: in frame {EML4:ALK}                                      | NA                           |
| EVT-P-0014<br>310-T02-IM<br>6-234 | P-0014<br>310-T0<br>2-IM6 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294475<br>44       | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+6128:EML4_c.3172+783:ALKinv   Protein Fusion: in frame {EML4:ALK}                                      | NA                           |
| EVT-P-0043<br>041-T01-IM<br>6-235 | P-0043<br>041-T0<br>1-IM6 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294465<br>97       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1782:EML4_c.3173-203:ALKinv   Protein Fusion: in frame {EML4:ALK}                                     | NA                           |
| EVT-P-0013<br>299-T02-IM<br>6-236 | P-0013<br>299-T0<br>2-IM6 |            | EML<br>4-ALK | EML<br>4     | in-frame     | three_prime | 294470<br>38       | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein<br>kinase<br>domain |              |                   | EML4 (NM_019063) -ALK (NM_004304)<br>Fusion(EML4 exon5 fused with ALK exon20) : c.667+1302:EML4_c.3173-644:ALKinv   Protein Fusion: in frame {EML4:ALK} | NA                           |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_nicated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                         | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|-----------------------|---------------------------------|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0012<br>913-T02-IM<br>6-237 | P-0012<br>913-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294469<br>65          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.209<br>-4133:EML4_c.3173-571:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0043<br>350-T01-IM<br>6-238 | P-0043<br>350-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294475<br>13          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-4017:EML4_c.3172+814:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0043<br>363-T01-IM<br>6-239 | P-0043<br>363-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294481<br>01          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.225<br>4:EML4_c.3172+226:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                   | NA                            |
| EVT-P-0043<br>594-T01-IM<br>6-240 | P-0043<br>594-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>81          | 19               | 1040                         | False                    | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.339<br>-2119:EML4_c.3118:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                   | NA                            |
| EVT-P-0069<br>396-T02-IM<br>7-241 | P-0069<br>396-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294469<br>44          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-4308:EML4_c.3173-550:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0069<br>535-T01-IM<br>7-242 | P-0069<br>535-T0<br>1-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294477<br>64          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.208<br>+3829:EML4_c.3172+563:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0069<br>617-T01-IM<br>7-243 | P-0069<br>617-T0<br>1-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294480<br>17          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+7821:EML4_c.3172+310:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0044<br>432-T01-IM<br>6-244 | P-0044<br>432-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>44          | 19               | 1052                         | False                    | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.205<br>7-576:EML4_c.3155:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                   | NA                            |
| EVT-P-0005<br>743-T02-IM<br>5-245 | P-0005<br>743-T0<br>2-IM5 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294476<br>69          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-20 with<br>ALK exons 20-29) : c.2243-<br>181:EML4_c.3172+658:AL<br>Kinv   Protein fusion: in<br>frame (EML4-ALK) | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status    | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|------------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0045<br>222-T03-IM<br>7-246 | P-0045<br>222-T0<br>3-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n  | thr<br>ee<br>_pr<br>ime | 294475<br>83          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.227<br>7:EML4_c.3172+744:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}          | NA                            |
| EVT-P-0045<br>243-T01-IM<br>6-247 | P-0045<br>243-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294482<br>98          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-3064:EML4_c.3172+29:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}       | NA                            |
| EVT-P-0045<br>341-T01-IM<br>6-248 | P-0045<br>341-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294479<br>20          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-4434:EML4_c.3172+407:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0065<br>469-T02-IM<br>7-249 | P-0065<br>469-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294465<br>65          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-1949:EML4_c.3173-171:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0065<br>210-T01-IM<br>7-250 | P-0065<br>210-T0<br>1-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294480<br>76          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-2012:EML4_c.3172+251:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0012<br>873-T03-IM<br>7-251 | P-0012<br>873-T0<br>3-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294464<br>39          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+2231:EML4_c.3173-45:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0064<br>716-T02-IM<br>7-252 | P-0064<br>716-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294477<br>39          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-3669:EML4_c.3172+588:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}      | NA                            |
| EVT-P-0031<br>368-T04-IM<br>6-253 | P-0031<br>368-T0<br>4-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294471<br>91          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-1154:EML4_c.3173-797:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                            |
| EVT-P-0045<br>912-T02-IM<br>6-254 | P-0045<br>912-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | out-of-fr<br>ame | thr<br>ee<br>_pr<br>ime | 294465<br>07          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.215<br>4+1011:EML4_c.3173-113:<br>ALKinv   Protein Fusion:<br>out of frame {EML4:ALK} | NA                            |
| EVT-P-0031<br>824-T01-IM<br>6-255 | P-0031<br>824-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr<br>ee<br>_pr<br>ime | 294505<br>57          | 17               | 939                          | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.208<br>+2971:EML4_c.2816-19:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}       | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_nicated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                            | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|-----------------------|---------------------------------|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0046<br>083-T02-IM<br>7-256 | P-0046<br>083-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294466<br>65          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+986:EML4_c.3173-271:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                            |
| EVT-P-0046<br>594-T01-IM<br>6-257 | P-0046<br>594-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294464<br>45          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+2217:EML4_c.3173-51:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0040<br>732-T02-IM<br>6-258 | P-0040<br>732-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294466<br>36          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-1832:EML4_c.3173-242:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                            |
| EVT-P-0047<br>355-T01-IM<br>6-259 | P-0047<br>355-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294466<br>09          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+925:EML4_c.3173-215:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0047<br>355-T02-IM<br>6-260 | P-0047<br>355-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294466<br>09          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+925:EML4_c.3173-215:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0064<br>342-T01-IM<br>7-261 | P-0064<br>342-T0<br>1-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>34          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.208<br>+4042:EML4_c.3173-140:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                            |
| EVT-P-0063<br>585-T01-IM<br>7-262 | P-0063<br>585-T0<br>1-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294468<br>84          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-2813:EML4_c.3173-490:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                            |
| EVT-P-0012<br>873-T02-IM<br>6-263 | P-0012<br>873-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294464<br>39          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+2231:EML4_c.3173-45:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                            |
| EVT-P-0062<br>999-T02-IM<br>7-264 | P-0062<br>999-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>91          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2055:EML4_c.3173-697:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK} | NA                            |
| EVT-P-0049<br>318-T01-IM<br>6-265 | P-0049<br>318-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294463<br>59          | 20               | 1070                         | False                    | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.215<br>4+2335:EML4_c.3208:ALKi<br>nv   Protein Fusion:<br>mid-exon {EML4:ALK}     | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name     | part_ner_gene | fra_me_status    | tar_get_role         | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_tru_nicated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                             | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|------------------|---------------|------------------|----------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|-----------------------|---------------------------------|---------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0046<br>720-T01-IM<br>6-266 | P-0046<br>720-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | out-of-fr<br>ame | thr_ee<br>_pr<br>ime | 294466<br>69          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.164<br>1+1:EML4_c.3173-275:AL<br>Kinv   Protein Fusion: out of<br>frame {EML4:ALK} | NA                            |
| EVT-P-0049<br>375-T02-IM<br>6-267 | P-0049<br>375-T0<br>2-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294467<br>32          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+1166:EML4_c.3173-338:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0049<br>511-T01-IM<br>6-268 | P-0049<br>511-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294469<br>91          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.224<br>2+9:EML4_c.3173-597:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}     | NA                            |
| EVT-P-0010<br>177-T02-IM<br>7-269 | P-0010<br>177-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294473<br>58          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-2649:EML4_c.3173-964:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}  | NA                            |
| EVT-P-0031<br>827-T03-IM<br>6-270 | P-0031<br>827-T0<br>3-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294466<br>46          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-639:EML4_c.3173-252:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0051<br>231-T01-IM<br>6-271 | P-0051<br>231-T0<br>1-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294467<br>74          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4130:EML4_c.3173-380:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0051<br>231-T03-IM<br>6-272 | P-0051<br>231-T0<br>3-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294467<br>74          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4130:EML4_c.3173-380:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0062<br>985-T02-IM<br>7-273 | P-0062<br>985-T0<br>2-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294471<br>99          | 20               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.667<br>+4561:EML4_c.3173-805:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}   | NA                            |
| EVT-P-0052<br>035-T03-IM<br>6-274 | P-0052<br>035-T0<br>3-IM6 |              | EML<br>4-AL<br>K | EML<br>4      | in-fr<br>ame     | thr_ee<br>_pr<br>ime | 294476<br>56          | 19               | 1058                         | True                     | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.224<br>2+79:EML4_c.3172+671:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}    | NA                            |
| EVT-P-0061<br>297-T03-IM<br>7-275 | P-0061<br>297-T0<br>3-IM7 |              | EML<br>4-AL<br>K | EML<br>4      | unk<br>now<br>n  | thr_ee<br>_pr<br>ime | 294483<br>75          | 19               | 1042                         | False                    | retain<br>ed   | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                    | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+931:EML4_c.3124:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}       | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_j_d | fusio_n_name     | part_ner_gen_e | fra_me_sta_tus  | tar_get_ro_le           | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_sta_tus | domain_r_etained_fr_action | domain_is_tru_nicated | retain_ed_do_mains              | lost_do_mains | disrupt_ed_do_mains | source_annotation_text                                                                                                                                                        | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|---------------|------------------|----------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|-----------------|----------------------------|-----------------------|---------------------------------|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0052<br>297-T02-IM<br>6-276 | P-0052<br>297-T0<br>2-IM6 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294482<br>55          | 19               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-161:EML4_c.3172+72:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0061<br>297-T02-IM<br>7-277 | P-0061<br>297-T0<br>2-IM7 |               | EML<br>4-AL<br>K | EML<br>4       | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>75          | 19               | 1042                         | False                    | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+931:EML4_c.3124:ALKin<br>v   Protein Fusion:<br>mid-exon {EML4:ALK}                                                  | NA                            |
| EVT-P-0005<br>786-T01-IM<br>5-278 | P-0005<br>786-T0<br>1-IM5 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294482<br>39          | 19               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion<br>(EML4 exons 1-18 with<br>ALK exons 20-29) : c.2057-<br>186:EML4_c.3172+88:ALKi<br>nv   Protein fusion: in frame<br>(EML4-ALK) | NA                            |
| EVT-P-0060<br>148-T01-IM<br>7-279 | P-0060<br>148-T0<br>1-IM7 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294477<br>57          | 19               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.148<br>9+2600:EML4_c.3172+570<br>:ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                             | NA                            |
| EVT-P-0059<br>699-T03-IM<br>7-280 | P-0059<br>699-T0<br>3-IM7 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>67          | 19               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-1990:EML4_c.3172+860:<br>ALKinv   Protein Fusion: in<br>frame {EML4:ALK}                                             | NA                            |
| EVT-P-0031<br>827-T06-IM<br>7-281 | P-0031<br>827-T0<br>6-IM7 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294466<br>46          | 20               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.149<br>0-639:EML4_c.3173-252:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                              | NA                            |
| EVT-P-0054<br>263-T01-IM<br>6-282 | P-0054<br>263-T0<br>1-IM6 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>57          | 20               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-210:EML4_c.3173-163:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0059<br>257-T02-IM<br>7-283 | P-0059<br>257-T0<br>2-IM7 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294482<br>10          | 19               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.668<br>-982:EML4_c.3172+117:AL<br>Kinv   Protein Fusion: in<br>frame {EML4:ALK}                                               | NA                            |
| EVT-P-0054<br>408-T01-IM<br>6-284 | P-0054<br>408-T0<br>1-IM6 |               | EML<br>4-AL<br>K | EML<br>4       | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294469<br>02          | 20               | 1058                         | True                     | retai<br>ned    | 1.0                        | False                 | Protein<br>kinase<br>domai<br>n |               |                     | EML4 (NM_019063) - ALK<br>(NM_004304) fusion: c.224<br>3-262:EML4_c.3173-508:A<br>LKinv   Protein Fusion: in<br>frame {EML4:ALK}                                              | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                     | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008<br>332-T02-IM<br>6-285 | P-0008<br>332-T0<br>2-IM6 |            | EML4-ALK    | EML4         | unknown      | three_prime | 29446913           | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173-519:ALKdup   Antisense Fusion                                                                                         | NA                           |
| EVT-P-0058<br>599-T02-IM<br>7-286 | P-0058<br>599-T0<br>2-IM7 |            | EML4-ALK    | EML4         | in-frame     | three_prime | 29446415           | 20              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1345:EML4_c.3173-21:ALKinv   Protein Fusion: in frame {EML4:ALK}                                                                         | NA                           |
| EVT-P-0044<br>042-T01-IM<br>6-287 | P-0044<br>042-T0<br>1-IM6 |            | EML4-ALK    | EML4         | in-frame     | three_prime | 29447727           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1681:EML4_c.3172+600:ALKinv   Protein Fusion: in frame {EML4:ALK}                                                                         | NA                           |
| EVT-P-0019<br>560-T01-IM<br>6-288 | P-0019<br>560-T0<br>1-IM6 |            | EML4-ALK    | EML4         | out-of-frame | three_prime | 29447952           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968-2879:EML4_c.3172+375:ALKinv   Protein Fusion: out of frame {EML4:ALK}                             | NA                           |
| EVT-P-0044<br>526-T02-IM<br>6-289 | P-0044<br>526-T0<br>2-IM6 |            | FN1-ALK     | FN1          | in-frame     | three_prime | 29448533           | 19              | 1023                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | FN1 (NM_212482) - ALK (NM_004304) fusion: c.1819+332:FN1_c.3068-102:ALKdel   Protein Fusion: in frame {FN1:ALK}                                                                            | NA                           |
| EVT-P-0030<br>808-T01-IM<br>6-290 | P-0030<br>808-T0<br>1-IM6 |            | HMBOX1-ALK  | HMBOX1       | unknown      | three_prime | 29447713           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | HMBOX1 (NM_001135726) ALK (NM_004304) Rearrangement : t(2;8)(p23.3,p21.1)(chr2:g.29447713::chr8:g.28904937)   Protein Fusion: mid-exon {HMBOX1:ALK}                                        | NA                           |
| EVT-P-0062<br>804-T03-IM<br>7-291 | P-0062<br>804-T0<br>3-IM7 |            | KIF5B-ALK   | KIF5B        | in-frame     | three_prime | 29447791           | 19              | 1058                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447791::chr10:g.32304833)   Protein Fusion: in frame {KIF5B:ALK}                                                | NA                           |
| EVT-P-0005<br>711-T02-IM<br>6-292 | P-0005<br>711-T0<br>2-IM6 |            | KIF5B-ALK   | KIF5B        | in-frame     | three_prime | 29449548           | 18              | 1023                        | True                   | retained      | 1.0                      | False               | Protein kinase domain |              |                   | KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548::chr10:g.32304765)   Protein fusion: in frame (KIF5B-ALK) | NA                           |

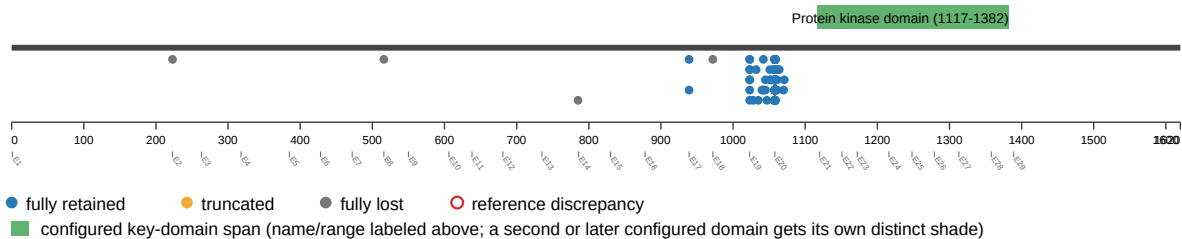
| event_id                          | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusio<br>n_n<br>ame | part<br>ner_<br>gene | fra<br>me_<br>sta<br>tus | tar<br>get_<br>ro<br>le  | breakp<br>oint_ge<br>nomic | break<br>point_<br>exon | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains        | lost_<br>do<br>mai<br>ns | disrupt<br>ed_do<br>mains | source_annotation_text                                                                                                                                                                     | source_site2<br>_effect_on_f<br>rame |
|-----------------------------------|---------------------------|----------------------------|---------------------|----------------------|--------------------------|--------------------------|----------------------------|-------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|---------------------------------|--------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-0008<br>764-T01-IM<br>5-293 | P-0008<br>764-T0<br>1-IM5 |                            | KIF5<br>B-AL<br>K   | KIF5<br>B            | unk<br>now<br>n          | thre<br>e_<br>_pr<br>ime | 294483<br>27               | 19                      | 1058                                | False                          | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750)   Protein fusion: mid-exon (KIF5B-ALK) | NA                                   |
| EVT-P-0017<br>924-T01-IM<br>6-294 | P-0017<br>924-T0<br>1-IM6 |                            | KIF5<br>B-AL<br>K   | KIF5<br>B            | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294479<br>45               | 19                      | 1058                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310)   Protein Fusion: in frame {KIF5B:ALK}                                                | NA                                   |
| EVT-P-0008<br>764-T02-IM<br>6-295 | P-0008<br>764-T0<br>2-IM6 |                            | KIF5<br>B-AL<br>K   | KIF5<br>B            | unk<br>now<br>n          | thre<br>e_<br>_pr<br>ime | 294483<br>27               | 19                      | 1058                                | False                          | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750)   Protein fusion: mid-exon (KIF5B-ALK) | NA                                   |
| EVT-P-0067<br>512-T01-IM<br>7-297 | P-0067<br>512-T0<br>1-IM7 |                            | NOT<br>CH1-<br>ALK  | NOT<br>CH1           | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294491<br>52               | 18                      | 1023                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | NOTCH1 (NM_017617) - ALK (NM_004304) fusion: t(2;9)(p23.2;q34.3)(chr2:g.29449152::chr9:g.139416748)   Protein Fusion: in frame {NOTCH1:ALK}                                                | NA                                   |
| EVT-P-0034<br>044-T01-IM<br>6-298 | P-0034<br>044-T0<br>1-IM6 |                            | NRP<br>2-AL<br>K    | NRP<br>2             | in-fr<br>ame             | thre<br>e_<br>_pr<br>ime | 294485<br>50               | 19                      | 1023                                | True                           | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv   Protein Fusion: in frame {NRP2:ALK}                                                                           | NA                                   |
| EVT-P-0055<br>667-T06-IM<br>7-299 | P-0055<br>667-T0<br>6-IM7 |                            | PICALM-<br>ALK      | PICALM               | unk<br>now<br>n          | thre<br>e_<br>_pr<br>ime | 294463<br>86               | 20                      | 1061                                | False                          | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391)   Protein Fusion: mid-exon {PICALM:ALK}                                               | NA                                   |
| EVT-P-0055<br>667-T02-IM<br>6-300 | P-0055<br>667-T0<br>2-IM6 |                            | PICALM-<br>ALK      | PICALM               | unk<br>now<br>n          | thre<br>e_<br>_pr<br>ime | 294463<br>86               | 20                      | 1061                                | False                          | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391)   Protein Fusion: mid-exon {PICALM:ALK}                                        | NA                                   |
| EVT-P-0055<br>667-T04-IM<br>6-301 | P-0055<br>667-T0<br>4-IM6 |                            | PICALM-<br>ALK      | PICALM               | unk<br>now<br>n          | thre<br>e_<br>_pr<br>ime | 294463<br>86               | 20                      | 1061                                | False                          | retai<br>ned              | 1.0                              | False                       | Protein<br>kinase<br>domai<br>n |                          |                           | PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391)   Protein Fusion: mid-exon {PICALM:ALK}                                               | NA                                   |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name            | part_ner_gene   | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                           | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|-------------------------|-----------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0054<br>253-T01-IM<br>6-302 | P-0054<br>253-T0<br>1-IM6 |              | PLE<br>KHA<br>7-AL<br>K | PLE<br>KHA<br>7 | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>96          | 19               | 1035                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | PLEKHA7 (NM_175058) -<br>ALK (NM_004304) fusion: t<br>(2;11)(p23.2;p15.1)(chr2:g.<br>29448396::chr11:g.168138<br>09)   Protein Fusion:<br>mid-exon {PLEKHA7:ALK} | NA                            |
| EVT-P-0059<br>884-T01-IM<br>7-303 | P-0059<br>884-T0<br>1-IM7 |              | PPP<br>1CB-<br>ALK      | PPP<br>1CB      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294474<br>41          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | PPP1CB (NM_206876) -<br>ALK (NM_004304) fusion:<br>c.52+5708:PPP1CB_c.317<br>2+886:ALKinv   Protein<br>Fusion: in frame<br>{PPP1CB:ALK}                          | NA                            |
| EVT-P-0048<br>893-T01-IM<br>6-307 | P-0048<br>893-T0<br>1-IM6 |              | SPT<br>BN1-<br>ALK      | SPT<br>BN1      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>46          | 19               | 1051                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | SPTBN1 (NM_003128) -<br>ALK (NM_004304) fusion:<br>c.763+1517:SPTBN1_c.31<br>53:ALKinv   Protein Fusion:<br>mid-exon {SPTBN1:ALK}                                | NA                            |
| EVT-P-0053<br>963-T02-IM<br>6-308 | P-0053<br>963-T0<br>2-IM6 |              | SPT<br>BN1-<br>ALK      | SPT<br>BN1      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>45          | 19               | 1052                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | SPTBN1 (NM_003128) -<br>ALK (NM_004304) fusion:<br>c.764-1123:SPTBN1_c.315<br>4:ALKinv   Protein Fusion:<br>mid-exon {SPTBN1:ALK}                                | NA                            |
| EVT-P-0045<br>724-T01-IM<br>6-309 | P-0045<br>724-T0<br>1-IM6 |              | SPT<br>BN1-<br>ALK      | SPT<br>BN1      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294467<br>96          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | SPTBN1 (NM_003128) -<br>ALK (NM_004304) fusion:<br>c.764-456:SPTBN1_c.3173<br>-402:ALKinv   Protein<br>Fusion: in frame<br>{SPTBN1:ALK}                          | NA                            |
| EVT-P-0042<br>596-T02-IM<br>6-310 | P-0042<br>596-T0<br>2-IM6 |              | SQS<br>TM1-<br>ALK      | SQS<br>TM1      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>11          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | SQSTM1 (NM_003900) -<br>ALK (NM_004304) fusion: t<br>(2;5)(p23.2;q35.3)(chr2:g.2<br>9446511::chr5:g.17925460<br>3)   Protein Fusion: in<br>frame {SQSTM1:ALK}    | NA                            |
| EVT-P-0042<br>596-T01-IM<br>6-311 | P-0042<br>596-T0<br>1-IM6 |              | SQS<br>TM1-<br>ALK      | SQS<br>TM1      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>11          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | SQSTM1 (NM_003900) -<br>ALK (NM_004304) fusion: t<br>(2;5)(p23.2;q35.3)(chr2:g.2<br>9446511::chr5:g.17925460<br>3)   Protein Fusion: in<br>frame {SQSTM1:ALK}    | NA                            |
| EVT-P-0053<br>901-T01-IM<br>6-312 | P-0053<br>901-T0<br>1-IM6 |              | STR<br>N-AL<br>K        | STR<br>N        | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294463<br>75          | 20               | 1064                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | STRN (NM_003162) - ALK<br>(NM_004304) fusion: c.339<br>-2565:STRN_c.3192:ALKd<br>el   Protein Fusion:<br>mid-exon {STRN:ALK}                                     | NA                            |
| EVT-P-0048<br>102-T03-IM<br>6-313 | P-0048<br>102-T0<br>3-IM6 |              | STR<br>N-AL<br>K        | STR<br>N        | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>56          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | STRN (NM_003162) - ALK<br>(NM_004304) fusion: c.412<br>+2761:STRN_c.3173-662:<br>ALKdel   Protein Fusion: in<br>frame {STRN:ALK}                                 | NA                            |

| event_id                          | sampl_e_id                | pa_tie_nt_id | fusio_n_name      | part_ner_gene | fra_me_status   | tar_get_role            | breakp_o_int_ge_nomic | break_point_exon | breakpoint_protein_p_osition | is_intron_ic_break_point | dom_ain_status | domain_r_etained_fr_action | domain_is_trun_cated | retain_ed_domains               | lost_do_mains | disrupt_ed_domains | source_annotation_text                                                                                                                                   | source_site2_effect_on_f_rame |
|-----------------------------------|---------------------------|--------------|-------------------|---------------|-----------------|-------------------------|-----------------------|------------------|------------------------------|--------------------------|----------------|----------------------------|----------------------|---------------------------------|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0033<br>066-T01-IM<br>6-314 | P-0033<br>066-T0<br>1-IM6 |              | STR<br>N-AL<br>K  | STR<br>N      | unk<br>now<br>n | thr<br>ee<br>_pr<br>ime | 294483<br>65          | 19               | 1045                         | False                    | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | STRN (NM_003162) - ALK (NM_004304) fusion( STRN exons 1-3 fused to ALK exons 19-29): c.413-1084:STRN_c.3134:ALKdel   Protein Fusion: mid-exon {STRN:ALK} | NA                            |
| EVT-P-0048<br>102-T02-IM<br>6-315 | P-0048<br>102-T0<br>2-IM6 |              | STR<br>N-AL<br>K  | STR<br>N      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294470<br>56          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662: ALKdel   Protein Fusion: in frame {STRN:ALK}                                      | NA                            |
| EVT-P-0056<br>097-T01-IM<br>6-316 | P-0056<br>097-T0<br>1-IM6 |              | TFG-<br>ALK       | TFG           | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294478<br>96          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | TFG (NM_001195478) - ALK (NM_004304) rearrangement: t(2;3)(p23.2;q12.2)(chr2:g.29447896::chr3:g.100456163)   Protein Fusion: in frame {TFG:ALK}          | NA                            |
| EVT-P-0045<br>363-T02-IM<br>6-318 | P-0045<br>363-T0<br>2-IM6 |              | TPM<br>1-AL<br>K  | TPM<br>1      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>14          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833)   Protein Fusion: in frame {TPM1:ALK}             | NA                            |
| EVT-P-0045<br>363-T01-IM<br>6-319 | P-0045<br>363-T0<br>1-IM6 |              | TPM<br>1-AL<br>K  | TPM<br>1      | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294465<br>14          | 20               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833)   Protein Fusion: in frame {TPM1:ALK}             | NA                            |
| EVT-P-0053<br>383-T01-IM<br>6-320 | P-0053<br>383-T0<br>1-IM6 |              | TRA<br>F3-A<br>LK | TRA<br>F3     | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294476<br>88          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | TRAF3 (NM_003300) - ALK (NM_004304) fusion: t (2;14)(p23.2;q32.32)(chr2:g.29447688::chr14:g.103370282)   Protein Fusion: in frame {TRAF3:ALK}            | NA                            |
| EVT-P-0001<br>737-T01-IM<br>3-321 | P-0001<br>737-T0<br>1-IM3 |              | VCL-<br>ALK       | VCL           | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294473<br>90          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | VCL (NM_014000) - ALK (NM_004304) Translocation : t(10;2)(q22.2;p23.2)(chr10:c.2435-486::chr2:c.3172+937)   Protein fusion: in frame (VCL-ALK)           | NA                            |
| EVT-P-0045<br>566-T01-IM<br>6-322 | P-0045<br>566-T0<br>1-IM6 |              | ZFP<br>M2-<br>ALK | ZFP<br>M2     | in-fr<br>ame    | thr<br>ee<br>_pr<br>ime | 294479<br>02          | 19               | 1058                         | True                     | retai<br>ned   | 1.0                        | False                | Protein<br>kinase<br>domai<br>n |               |                    | ZFPM2 (NM_012082) - ALK (NM_004304) fusion: t (2;8)(p23.2;q23.1)(chr2:g.29447902::chr8:g.106384092)   Protein Fusion: in frame {ZFPM2:ALK}               | NA                            |

## domain retention outliers

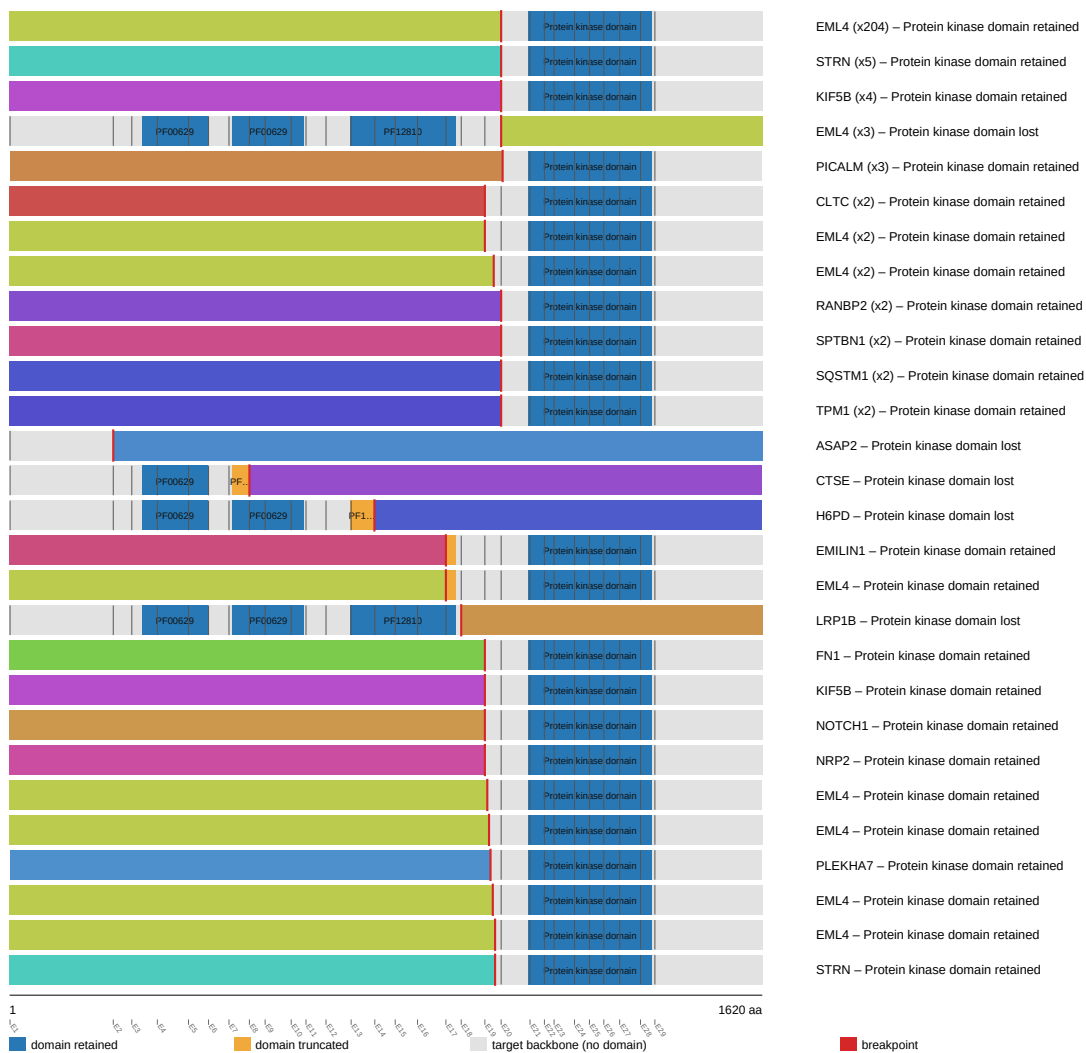
### ALK domain-retention track



## fusion schematic

ALK fusion-transcript schematic

partner block → breakpoint → retained ALK portion (domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

# intragenic deletion schematic



## reference comparison

Reference vs msk\_impact\_50k\_2026

